H4 Comprehensive Variable Audit

Step 1: Complete column inventory and analytical roles.
Step 2: Existing sensor consolidation and exposure-representation rules, unchanged.
Step 3: Measurement provenance and shared-design diagnostics; all Step-2 eligible fields remain candidates.
No VIF elimination stage. Current fitted equation and diagnostics are in H4_Step5_Equations_Checks.txt.

Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
1	Empatica	empatica_eda_phasic	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
2	Empatica	empatica_eda_tonic	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
3	Empatica	empatica_heart_rate	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
4	Empatica	empatica_hrv_fd_hf	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
5	Empatica	empatica_hrv_fd_hfnu	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
6	Empatica	empatica_hrv_fd_lf	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
7	Empatica	empatica_hrv_fd_lf_hf_ratio	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
8	Empatica	empatica_hrv_fd_lfnu	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
9	Empatica	empatica_hrv_fd_total_power	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
10	Empatica	empatica_hrv_fd_vlf	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
11	Empatica	empatica_hrv_td_rmssd	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
12	Empatica	empatica_hrv_td_sdnn	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
13	Empatica	empatica_skin_temperature	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
14	Eye-Tracker	eyetracker_blink_duration_s	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
15	Eye-Tracker	eyetracker_distance_from_center	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome

H4 Comprehensive Variable Audit

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Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
16	Eye-Tracker	eyetracker_eyelid_aperture_avg	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
17	Eye-Tracker	eyetracker_fixation_duration_s	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
18	Eye-Tracker	eyetracker_fixation_rate	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
19	Eye-Tracker	eyetracker_gaze_centrality	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
20	Eye-Tracker	eyetracker_gaze_dispersion	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
21	Eye-Tracker	eyetracker_gaze_velocity	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
22	Eye-Tracker	eyetracker_pupil_change_rate	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
23	Eye-Tracker	eyetracker_pupil_diameter_avg	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
24	Eye-Tracker	eyetracker_saccade_amplitude	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
25	Eye-Tracker	eyetracker_saccade_duration_s	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
26	Eye-Tracker	eyetracker_saccade_peak_velocity	Outcome Y - Physiological	Analyzed as outcome Y	-	Physiological outcome
27	Questionnaire	questionnaire_current_comfort_1_7	Outcome Y - Psychological	Analyzed as outcome Y	-	Psychological outcome
28	Questionnaire	questionnaire_pleasantness_1_5	Outcome Y - Psychological	Analyzed as outcome Y	-	Psychological outcome
29	Questionnaire	questionnaire_prs11_total_mean_1_5	Outcome Y - Psychological	Analyzed as outcome Y	-	Psychological outcome
30	Questionnaire	questionnaire_stai6_mean_1_5	Outcome Y - Psychological	Analyzed as outcome Y	-	Psychological outcome

H4 Comprehensive Variable Audit

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Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
31	Acoustic Spectral	Octave Frequency Bands (9 variables)	Environmental Candidate (X)	Not advanced	-	EXPLORATORY ACOUSTIC-FREQUENCY CONSTRUCT
32	Acoustic Spectral	dominant_frequency_hz	Environmental Candidate (X)	Not advanced	-	EXPLORATORY ACOUSTIC-FREQUENCY CONSTRUCT
33	Acoustic Spectral	freq_prop_1000_4000	Environmental Candidate (X)	Not advanced	-	EXPLORATORY ACOUSTIC-FREQUENCY CONSTRUCT
34	Acoustic Spectral	freq_prop_125_1000	Environmental Candidate (X)	Not advanced	-	EXPLORATORY ACOUSTIC-FREQUENCY CONSTRUCT
35	Acoustic Spectral	freq_prop_20_125	Environmental Candidate (X)	Not advanced	-	EXPLORATORY ACOUSTIC-FREQUENCY CONSTRUCT
36	Acoustic Spectral	freq_prop_4000_8000	Environmental Candidate (X)	Not advanced	-	EXPLORATORY ACOUSTIC-FREQUENCY CONSTRUCT
37	Acoustic Spectral	freq_prop_8000_20000	Environmental Candidate (X)	Not advanced	-	EXPLORATORY ACOUSTIC-FREQUENCY CONSTRUCT
38	Acoustic Spectral	spectral_centroid_hz	Environmental Candidate (X)	Not advanced	-	EXPLORATORY ACOUSTIC-FREQUENCY CONSTRUCT
39	Acoustic Spectral	spectral_rolloff85_hz	Environmental Candidate (X)	Not advanced	-	EXPLORATORY ACOUSTIC-FREQUENCY CONSTRUCT
40	Atmotube	atmo_lys_LYS1__lys_b'	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)
41	Atmotube	atmo_lys_LYS1__lys_g'	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)
42	Atmotube	atmo_lys_LYS1__lys_kelvin	Environmental Candidate (X)	Not advanced	-	CONSOLIDATED RAW STREAM (EXCLUDED)
43	Atmotube	atmo_lys_LYS1__lys_lux	Environmental Candidate (X)	Not advanced	-	CONSOLIDATED RAW STREAM (EXCLUDED)
44	Atmotube	atmo_lys_LYS1__lys_medi	Environmental Candidate (X)	Not advanced	-	CONSOLIDATED RAW STREAM (EXCLUDED)
45	Atmotube	atmo_lys_LYS1__lys_r'	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)

H4 Comprehensive Variable Audit

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Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
46	Atmotube	atmo_lys_LYS1__lys_rgbb	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)
47	Atmotube	atmo_lys_LYS1__lys_rgbg	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)
48	Atmotube	atmo_lys_LYS1__lys_rgbir	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)
49	Atmotube	atmo_lys_LYS1__lys_rgbr	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)
50	Atmotube	atmo_lys_LYS2__lys_b'	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)
51	Atmotube	atmo_lys_LYS2__lys_g'	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)
52	Atmotube	atmo_lys_LYS2__lys_kelvin	Environmental Candidate (X)	Not advanced	-	CONSOLIDATED RAW STREAM (EXCLUDED)
53	Atmotube	atmo_lys_LYS2__lys_lux	Environmental Candidate (X)	Not advanced	-	CONSOLIDATED RAW STREAM (EXCLUDED)
54	Atmotube	atmo_lys_LYS2__lys_medi	Environmental Candidate (X)	Not advanced	-	CONSOLIDATED RAW STREAM (EXCLUDED)
55	Atmotube	atmo_lys_LYS2__lys_r'	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)
56	Atmotube	atmo_lys_LYS2__lys_rgbb	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)
57	Atmotube	atmo_lys_LYS2__lys_rgbg	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)
58	Atmotube	atmo_lys_LYS2__lys_rgbir	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)
59	Atmotube	atmo_lys_LYS2__lys_rgbr	Environmental Candidate (X)	Not advanced	-	AUXILIARY DEVICE CHANNEL (EXCLUDED)
60	Atmotube	atmo_lys_atmotube_left__atmotube_humidity	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)

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Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
61	Atmotube	atmo_lys_atmotube_left__atmotube_pm1	Environmental Candidate (X)	Not advanced	-	CONSOLIDATED RAW STREAM (EXCLUDED)
62	Atmotube	atmo_lys_atmotube_left__atmotube_pm10	Environmental Candidate (X)	Not advanced	-	CONSOLIDATED RAW STREAM (EXCLUDED)
63	Atmotube	atmo_lys_atmotube_left__atmotube_pm2.5	Environmental Candidate (X)	Not advanced	-	CONSOLIDATED RAW STREAM (EXCLUDED)
64	Atmotube	atmo_lys_atmotube_left__atmotube_temperature	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
65	Atmotube	atmo_lys_atmotube_right__atmotube_humidity	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
66	Atmotube	atmo_lys_atmotube_right__atmotube_pm1	Environmental Candidate (X)	Not advanced	-	CONSOLIDATED RAW STREAM (EXCLUDED)
67	Atmotube	atmo_lys_atmotube_right__atmotube_pm10	Environmental Candidate (X)	Not advanced	-	CONSOLIDATED RAW STREAM (EXCLUDED)
68	Atmotube	atmo_lys_atmotube_right__atmotube_pm2.5	Environmental Candidate (X)	Not advanced	-	CONSOLIDATED RAW STREAM (EXCLUDED)
69	Atmotube	atmo_lys_atmotube_right__atmotube_temperature	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
70	Atmotube	atmo_pm10_avg	Environmental Candidate (X)	Eligible	A Averaged instrument measurement	Candidate; model specification pending
71	Atmotube	atmo_pm1_avg	Environmental Candidate (X)	Eligible	A Averaged instrument measurement	Candidate; model specification pending
72	Atmotube	atmo_pm2.5_avg	Environmental Candidate (X)	Eligible	A Averaged instrument measurement	Candidate; model specification pending
73	LYS Button	lys_kelvin_avg	Environmental Candidate (X)	Eligible	D Derived metric	Candidate; model specification pending
74	LYS Button	lys_lux_avg	Environmental Candidate (X)	Eligible	A Averaged instrument measurement	Candidate; model specification pending
75	LYS Button	lys_medi_avg	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)

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Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
76	Solar Models	Clearness_Index	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
77	Solar Models	Clearness_Index_G0	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
78	Solar Models	Clearness_Index_Infer	Environmental Candidate (X)	Not advanced	-	NON-NUMERIC LABEL (EXCLUDED)
79	Street View / GIS	Building_%	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
80	Street View / GIS	Combined_Vegetation_%	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
81	Street View / GIS	Grass_%	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
82	Street View / GIS	Plant_%	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
83	Street View / GIS	SVF_Holmer	Environmental Candidate (X)	Eligible	D Image-derived sky-view factor	Candidate; model specification pending
84	Street View / GIS	Sky_%	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
85	Street View / GIS	Tree_%	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
86	Street View / GIS	Wall_%	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
87	Street View / GIS	grass_phase_mean	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
88	Street View / GIS	gvi_50m	Environmental Candidate (X)	Eligible	D Derived metric	Candidate; model specification pending
89	Street View / GIS	gvi_grass_50m	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
90	Street View / GIS	gvi_phase_mean	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)

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Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
91	Street View / GIS	gvi_plant_50m	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
92	Street View / GIS	gvi_tree_50m	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
93	Street View / GIS	plant_phase_mean	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
94	Street View / GIS	tree_phase_mean	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
95	UCM Backpack	ucm_AH	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
96	UCM Backpack	ucm_AIR_RH	Environmental Candidate (X)	Eligible	M Instrument-reported measurement	Candidate; model specification pending
97	UCM Backpack	ucm_AIR_T_bot	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
98	UCM Backpack	ucm_AIR_T_mid	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
99	UCM Backpack	ucm_AIR_T_top	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
100	UCM Backpack	ucm_AIR_temp	Environmental Candidate (X)	Eligible	M Instrument-reported measurement	Candidate; model specification pending
101	UCM Backpack	ucm_HUMIDEX	Environmental Candidate (X)	Eligible	D Derived metric	Candidate; model specification pending
102	UCM Backpack	ucm_HUMIDEX_SR	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
103	UCM Backpack	ucm_IO_flag	Environmental Candidate (X)	Not advanced	-	DISCRETE PROTOCOL FLAG (EXCLUDED)
104	UCM Backpack	ucm_IR_back	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
105	UCM Backpack	ucm_IR_down	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)

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Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
106	UCM Backpack	ucm_IR_front	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
107	UCM Backpack	ucm_IR_left	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
108	UCM Backpack	ucm_IR_right	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
109	UCM Backpack	ucm_IR_spot_gnd	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
110	UCM Backpack	ucm_IR_spot_left	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
111	UCM Backpack	ucm_IR_spot_right	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
112	UCM Backpack	ucm_IR_spot_sky	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
113	UCM Backpack	ucm_IR_up	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
114	UCM Backpack	ucm_MRT	Environmental Candidate (X)	Eligible	D Derived metric	Candidate; model specification pending
115	UCM Backpack	ucm_MRT_S	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
116	UCM Backpack	ucm_OPT	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
117	UCM Backpack	ucm_PET	Environmental Candidate (X)	Eligible	D Derived metric	Candidate; model specification pending
118	UCM Backpack	ucm_PET_SR	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
119	UCM Backpack	ucm_SND_dBA	Environmental Candidate (X)	Eligible	M Instrument-reported measurement	Candidate; model specification pending
120	UCM Backpack	ucm_SUN_Gh	Environmental Candidate (X)	Eligible	M Instrument-reported measurement	Candidate; model specification pending

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Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
121	UCM Backpack	ucm_SUN_alt	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
122	UCM Backpack	ucm_SUN_az	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
123	UCM Backpack	ucm_UTCI	Environmental Candidate (X)	Eligible	D Derived metric	Candidate; model specification pending
124	UCM Backpack	ucm_UTCI_SR	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
125	UCM Backpack	ucm_WIND_AWA	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
126	UCM Backpack	ucm_WIND_AWS	Environmental Candidate (X)	Eligible	M Instrument-reported apparent wind speed (moving platform)	Candidate; model specification pending
127	UCM Backpack	ucm_WIND_TWD	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
128	UCM Backpack	ucm_WIND_TWS	Environmental Candidate (X)	Not advanced	-	ALTERNATIVE REPRESENTATION (EXCLUDED)
129	UCM Backpack	ucm_fp	Environmental Candidate (X)	Not advanced	-	UNVERIFIED DEFINITION (EXCLUDED)
130	YAMNet Audio	category	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
131	YAMNet Audio	mechanisms_prop	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
132	YAMNet Audio	mechanisms_score	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
133	YAMNet Audio	rural_natural_prop	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
134	YAMNet Audio	rural_natural_score	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
135	YAMNet Audio	unidentify_prop	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT

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Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
136	YAMNet Audio	urban_manmade_negative_prop	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
137	YAMNet Audio	urban_manmade_negative_score	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
138	YAMNet Audio	urban_manmade_neutral_prop	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
139	YAMNet Audio	urban_manmade_neutral_score	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
140	YAMNet Audio	urban_manmade_positive_prop	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
141	YAMNet Audio	urban_manmade_positive_score	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
142	YAMNet Audio	vehicle_prop	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
143	YAMNet Audio	vehicle_score	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
144	YAMNet Audio	wind_prop	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
145	YAMNet Audio	wind_score	Environmental Candidate (X)	Not advanced	-	EXPLORATORY SOUNDSCAPE CONSTRUCT
146	Empatica	empatica__met	Movement Covariate (Z)	Exertion covariate proxy	-	Movement covariate
147	Empatica	empatica_acc_x	Movement Covariate (Z)	Exertion covariate proxy	-	Movement covariate
148	Empatica	empatica_acc_y	Movement Covariate (Z)	Exertion covariate proxy	-	Movement covariate
149	Empatica	empatica_acc_z	Movement Covariate (Z)	Exertion covariate proxy	-	Movement covariate
150	Empatica	empatica_vector_magnitude	Movement Covariate (Z)	Exertion covariate proxy	-	Movement covariate

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Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
151	LYS Button	atmo_lys_LYS1__lys_movement	Movement Covariate (Z)	Exertion covariate proxy	-	Movement covariate
152	LYS Button	atmo_lys_LYS2__lys_movement	Movement Covariate (Z)	Exertion covariate proxy	-	Movement covariate
153	UCM Backpack	ucm_GPS_speed_kmh	Movement Covariate (Z)	Exertion covariate proxy	-	Movement covariate
154	Atmotube	atmo_lys_Date	Metadata / QC - Not Candidate	-	-	STUDY METADATA
155	Empatica	empatica__eda_scl_usiemens	Metadata / QC - Not Candidate	-	-	QC METRIC
156	Empatica	empatica_hrv_fd_5min_available	Metadata / QC - Not Candidate	-	-	QC METRIC
157	Empatica	empatica_hrv_fd_5min_max_gap_seconds	Metadata / QC - Not Candidate	-	-	QC METRIC
158	Empatica	empatica_hrv_fd_5min_removed_pct	Metadata / QC - Not Candidate	-	-	QC METRIC
159	Empatica	empatica_hrv_fd_5min_segment_end	Metadata / QC - Not Candidate	-	-	QC METRIC
160	Empatica	empatica_hrv_fd_5min_segment_start	Metadata / QC - Not Candidate	-	-	QC METRIC
161	Empatica	empatica_hrv_fd_5min_valid_rri_count	Metadata / QC - Not Candidate	-	-	QC METRIC
162	Empatica	empatica_hrv_fd_5min_valid_span_seconds	Metadata / QC - Not Candidate	-	-	QC METRIC
163	Eye-Tracker	eyetracker_Date	Metadata / QC - Not Candidate	-	-	STUDY METADATA
164	Eye-Tracker	eyetracker_stress_composite	Metadata / QC - Not Candidate	-	-	QC METRIC
165	LYS Button	atmo_lys_LYS1__lux_cat	Outside Declared H4 Set	-	-	DERIVED LIGHT CATEGORY

H4 Comprehensive Variable Audit

Step 1: Complete column inventory and analytical roles.
Step 2: Existing sensor consolidation and exposure-representation rules, unchanged.
Step 3: Measurement provenance and shared-design diagnostics; all Step-2 eligible fields remain candidates.
No VIF elimination stage. Current fitted equation and diagnostics are in H4_Step5_Equations_Checks.txt.

Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
166	LYS Button	atmo_lys_LYS2__lux_cat	Outside Declared H4 Set	-	-	DERIVED LIGHT CATEGORY
167	Questionnaire	questionnaire_background_age	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
168	Questionnaire	questionnaire_background_bike_frequency	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
169	Questionnaire	questionnaire_background_commute_modes	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
170	Questionnaire	questionnaire_background_daily_walking	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
171	Questionnaire	questionnaire_background_gender	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
172	Questionnaire	questionnaire_background_has_children	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
173	Questionnaire	questionnaire_background_name	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
174	Questionnaire	questionnaire_background_nationality	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
175	Questionnaire	questionnaire_background_own_bike_for_experiment	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
176	Questionnaire	questionnaire_background_regular_physical_activity	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
177	Questionnaire	questionnaire_background_residence_type	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
178	Questionnaire	questionnaire_background_residential_area	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
179	Questionnaire	questionnaire_background_site_familiarity_1_10	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
180	Questionnaire	questionnaire_background_vision_correction	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA

H4 Comprehensive Variable Audit

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No VIF elimination stage. Current fitted equation and diagnostics are in H4_Step5_Equations_Checks.txt.

Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
181	Questionnaire	questionnaire_background_years_in_brussels	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
182	Questionnaire	questionnaire_bmi_category_who	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
183	Questionnaire	questionnaire_bmi_kg_m2	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
184	Questionnaire	questionnaire_clothing_accessories	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
185	Questionnaire	questionnaire_clothing_footwear	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
186	Questionnaire	questionnaire_clothing_height_raw	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
187	Questionnaire	questionnaire_clothing_lower_body	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
188	Questionnaire	questionnaire_clothing_skin_tone	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
189	Questionnaire	questionnaire_clothing_upper_body	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
190	Questionnaire	questionnaire_clothing_weight_raw	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
191	Questionnaire	questionnaire_environmental_comfort_mean_1_5	Metadata / QC - Not Candidate	-	-	QC METRIC
192	Questionnaire	questionnaire_height_m	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
193	Questionnaire	questionnaire_participant_id	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
194	Questionnaire	questionnaire_prs11_being_away_mean_1_5	Metadata / QC - Not Candidate	-	-	QC METRIC
195	Questionnaire	questionnaire_prs11_coherence_mean_1_5	Metadata / QC - Not Candidate	-	-	QC METRIC

H4 Comprehensive Variable Audit

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Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
196	Questionnaire	questionnaire_prs11_fascination_mean_1_5	Metadata / QC - Not Candidate	-	-	QC METRIC
197	Questionnaire	questionnaire_prs11_scope_mean_1_5	Metadata / QC - Not Candidate	-	-	QC METRIC
198	Questionnaire	questionnaire_response_index_within_participant	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
199	Questionnaire	questionnaire_space_familiarity_1_5	Metadata / QC - Not Candidate	-	-	QC METRIC
200	Questionnaire	questionnaire_thermal_comfort_1_5	Metadata / QC - Not Candidate	-	-	QC METRIC
201	Questionnaire	questionnaire_weight_kg	Metadata / QC - Not Candidate	-	-	DEMOGRAPHIC METADATA
202	Street View / GIS	gvi_match_dist_m	Metadata / QC - Not Candidate	-	-	STUDY METADATA
203	Study Design	Date	Metadata / QC - Not Candidate	-	-	STUDY METADATA
204	Study Design	Datetime	Metadata / QC - Not Candidate	-	-	STUDY METADATA
205	Study Design	ParticipantID	Metadata / QC - Not Candidate	-	-	STUDY METADATA
206	Study Design	PhaseID	Metadata / QC - Not Candidate	-	-	STUDY METADATA
207	Study Design	pct_complete	Metadata / QC - Not Candidate	-	-	STUDY METADATA
208	UCM Backpack	ucm_Date	Metadata / QC - Not Candidate	-	-	STUDY METADATA
209	UCM Backpack	ucm_GPS_HDOP	Metadata / QC - Not Candidate	-	-	STUDY METADATA
210	UCM Backpack	ucm_GPS_alt	Metadata / QC - Not Candidate	-	-	STUDY METADATA

H4 Comprehensive Variable Audit

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Sr. No.	Data Source (Sensor)	Variable	Step 1: Role	Step 2: Preprocessing	Step 3: Measurement provenance	Final role
211	UCM Backpack	ucm_GPS_hdg	Metadata / QC - Not Candidate	-	-	STUDY METADATA
212	UCM Backpack	ucm_GPS_lat	Metadata / QC - Not Candidate	-	-	STUDY METADATA
213	UCM Backpack	ucm_GPS_lon	Metadata / QC - Not Candidate	-	-	STUDY METADATA
214	UCM Backpack	ucm_GPS_speed_valid_sample_count	Metadata / QC - Not Candidate	-	-	STUDY METADATA
215	UCM Backpack	ucm_MAG_hdg	Metadata / QC - Not Candidate	-	-	STUDY METADATA
216	YAMNet Audio	end_time_s	Metadata / QC - Not Candidate	-	-	QC METRIC
217	YAMNet Audio	phase_start_offset_s	Metadata / QC - Not Candidate	-	-	QC METRIC
218	YAMNet Audio	segment	Metadata / QC - Not Candidate	-	-	QC METRIC